

Rewritten Chain of Thought: Refuting the Single-Rational-Function Hint and Reaching Only a Chamber-Restricted A_n in the Two-Minus Sector

Rewritten Chain of Thought

Orienting myself. The object is the on-shell tree amplitude A_n in the sector $\sigma = (-1, -1, +1, \dots, +1)$: legs 1, 2 are “minus”, legs 3, $\dots$, n are “plus”. The dispersion ties momentum to frequency, $\omega_i^2 = g|k_i|$ with $k_i = \sigma_i \omega_i^2 / g$, and the resonant manifold imposes $\sum_i \omega_i = 0$ and $\sum_i \sigma_i \omega_i^2 = 0$. The recursion glues vertices with a propagator

$$\frac{-i}{\omega_S^2 / |k_S| - g}, \quad k_S = \sum_{i \in S} k_i = \frac{1}{g} \sum_{i \in S} \sigma_i \omega_i^2.$$

The hint wants $A_n = N(\omega)/D(\omega)$ with D a product of channel factors, fit by matching at many points, a *single* global rational function with no $|\cdot|$, no min/max, no chambers, and not a plain polynomial. I keep that target in mind, but the $|k_S|$ in the propagator is an immediate red flag, so I resolve to test the global-rational claim rather than assume it.

First contact: amplitudes and the scaling weight. My first script just runs the code. At one generic point per n I get

$$A_5 = -\frac{2349}{4}i, \quad A_6 = -\frac{483327}{152}i, \quad A_7 = -\frac{7302393}{400}i,$$

with A_7 already costing about 19 seconds in exact arithmetic — a hint that I must be economical at high n . My $n = 4$ point returns **Indeterminate**; I note it and move on, suspecting an accidental degeneracy ($k_{13} = 0$ at the point I chose). I then scale all frequencies $\omega \rightarrow \lambda\omega$ and read off the weight: A_5 multiplies by 2^6 , A_6 by 2^8 . So

$$\deg A_n = 2n - 4.$$

Hypothesis 1, the bookkeeping one: $A_n = N/D$ is homogeneous of degree $2n - 4$ (counting $\deg \omega_i = 1$, since $k \sim \omega^2$ and the propagator is weight 0).

Thinking about the denominator from physics. Where are the poles? A propagator goes on shell at $\omega_S^2 = |P_S|$, where $P_S \equiv \sum_{i \in S} \sigma_i \omega_i^2$. External legs are safe, $|k_i| = \omega_i^2 / g$, because $\omega_i^2 \geq 0$. Internal lines are the worry. Splitting subsets by how they meet $\{1, 2\}$:

- all-plus $S \subseteq \{3, \dots, n\}$: $P_S = \sum_{i \in S} \omega_i^2 > 0$, definite, and the channel factor is the clean polynomial $\omega_S^2 - \sum_{i \in S} \omega_i^2 = 2e_2(S)$ with $e_2(S) = \sum_{i < j \in S} \omega_i \omega_j$;
- S meeting exactly one of $\{1, 2\}$: $P_S = -\omega_a^2 + (\text{plus})$, sign indefinite.

Hypothesis 2: maybe $D_n = \prod_{S \subseteq \{3..n\}, |S| \geq 2} 2e_2(S)$ (the safe channels only), and the mixed channels carry no pole. For $n = 5$ that would be $D_5 = (\omega_3\omega_4)(\omega_3\omega_5)(\omega_4\omega_5)(\omega_3\omega_4 + \omega_3\omega_5 + \omega_4\omega_5)$, degree 8, forcing $\deg N = 2n - 4 + 8 = 14$ — uncomfortably large, and it presupposes the mixed channels are inert. I distrust this enough to test the rationality claim directly before committing to any D .

The bottom of the tower forces my hand ($n = 4$). At $n = 4$ the two conservation laws on four frequencies are rigid. Taking ω_2, ω_3 free and solving, the solution is *forced*:

$$\omega_1 = -\omega_3, \quad \omega_4 = -\omega_2,$$

a two-parameter family $(-\omega_3, \omega_2, \omega_3, -\omega_2)$. (This also explains my earlier **Indeterminate**: with $\omega_1 = -\omega_3$ one gets $k_2 + k_4 = 0$ and $\omega_2 + \omega_4 = 0$, so a sub-current propagator is $0/0$.) Computing the amplitude symbolically and simplifying with the magnitudes kept symbolic gives a *piecewise* answer:

$$A_4 = -8i\omega_2^3\omega_3 \quad (\omega_2 < \omega_3), \quad A_4 = -8i\omega_2\omega_3^3 \quad (\omega_2 > \omega_3).$$

This is decisive and unwelcome: in signed variables the pieces are $-8i\omega_1\omega_4^3$ and $-8i\omega_1^3\omega_4$, switching across $\omega_2^2 = \omega_3^2$. By the identity theorem a single rational function equal to $-8i\omega_2^3\omega_3$ on an open set must equal it everywhere; it does not. So A_4 is genuinely piecewise. Hypothesis 3 (*the amplitude is globally rational*) is in serious doubt. I conjecture instead that the hint’s “single rational function” holds only chamber-by-chamber, and that the benchmark’s sampling lives in one chamber.

A method to see chambers: resolve $|\cdot|$ by a reference sign. To get an honest rational function I evaluate every $|x|$ in the symbolic amplitude by the sign of x at a chosen numeric reference point; that is exactly valid throughout the open sign-chamber containing the reference, and I can then **Together/Factor**. At $n = 5$ the raw amplitude has leafcount $\sim 4 \times 10^5$, but the chamber with free frequencies $(a, b, c) = (\omega_2, \omega_3, \omega_4) = (2, 3, 5)$ collapses to

$$A_5 = \frac{-16i a^5(ab + b^2 + ac + bc + c^2)}{a + b + c}.$$

Two other reference chambers, $(7, 11, 2)$ and $(3, 5, -2)$, give visibly *different* rational functions. Hypothesis 3 is now refuted outright: A_5 is chamber-dependent.

Mapping where this chamber holds. I test $f_A(a, b, c) = -16i a^5(ab + b^2 + ac + bc + c^2)/(a + b + c)$ against the recursion. It matches at $(2, 3, 5)$ and $(2, 5, 3)$ but not at $(3, 2, 5)$ (ratio $56/81$), $(5, 2, 3)$ (ratio $72/625$), $(3, 5, 2)$, $(5, 3, 2)$. Reading the pattern: f_A is exact precisely when $a = \omega_2$ is the *smallest* of a, b, c , and it is symmetric in $b \leftrightarrow c$ (i.e. in the two free plus legs). So the chamber is “ ω_2 smallest”.

The collapse to a monomial. The denominator $a + b + c$ is exactly the quantity that blows up in the kinematic solver, so it should be a coordinate artifact. Using the conservation laws I find the identities

$$ab + b^2 + ac + bc + c^2 = \omega_1(\omega_1 + \omega_5), \quad a + b + c = -(\omega_1 + \omega_5),$$

and the artifact cancels:

$$A_5|_{\omega_2 \text{ smallest}} = 16i\omega_1\omega_2^5.$$

A pure monomial. I check it head-on: $(2, 3, 5) \mapsto -3328i$, $(3, 5, 7) \mapsto -37584i$, $(1, 4, 9) \mapsto -1168i/7$ — all exact. As a function it uses only the two minus-leg frequencies, with the high power on the smaller one. This is beautiful but it is a *polynomial*, contradicting “not a plain polynomial”: another sign that the hint describes a per-chamber rational form, not the on-shell value.

Cataloguing the other chambers ($n = 5$). I grind out five more reference chambers and reduce each to ω variables:

$$\begin{aligned}\omega_2 \text{ smallest : } & 16i \omega_1 \omega_2^5, \\ \omega_3 \text{ smallest, } \omega_2 \text{ next : } & 16i \omega_1 \omega_2 \omega_3^2 (2\omega_2^2 - \omega_3^2), \\ \omega_3, \omega_4 \text{ smallest : } & 32i \omega_1 \omega_2 \omega_3^2 \omega_4^2, \\ \omega_5, \omega_3 \text{ smallest : } & 32i \omega_1 \omega_2 \omega_3^2 \omega_5^2, \\ \omega_5, \omega_2 \text{ smallest : } & 16i \omega_1 \omega_2 \omega_5^2 (2\omega_2^2 - \omega_5^2).\end{aligned}$$

(For the last two I needed an auxiliary identity: the quartic Q appearing there equals $a^2 s^2 - 2abc s - b^2 c^2$ with $s = a + b + c$, and $Q/s^2 = 2\omega_2^2 - \omega_5^2$ on shell.) Every chamber reduces to a *polynomial*; the solver artifact always cancels. So A_5 is a symmetric, piecewise-*polynomial* function whose pieces are labelled by the ordering of the $|\omega_i|$ — precisely the min / max structure the hint forbids. Hypothesis 4 (*some chamber has a genuine non-polynomial pole*) fails too: on shell there are no poles at all.

Symmetry check. $16i \omega_1 \omega_2^5$ looks asymmetric under $1 \leftrightarrow 2$, but the recursion picks leg 1 as a root, so I test directly: permuting the (k, ω) pairs — swap the minus legs, swap two plus legs, cycle $3 \rightarrow 4 \rightarrow 5$, or an arbitrary permutation $\{3, 1, 5, 2, 4\}$ — all return $-3328i$ at the $(2, 3, 5)$ point. So A_n is fully symmetric under same- σ relabelling; the monomial is a chamber *representative*, valid when ω_2 happens to be the smallest magnitude.

The conjecture and its test ($n = 4, 5, 6, 7$). The clean chamber generalises. The coefficients 8, 16, ... at $n = 4, 5$ suggest 2^{n-1} and the power $2n - 5$, so

$$\boxed{A_n = 2^{n-1} i \omega_1 \omega_2^{2n-5}} \quad \text{when } |\omega_2| = \min_i |\omega_i|.$$

I throw a spread of points at it. At $n = 5$: $(2, 3, 5), (1, 4, 9)$ exact. At $n = 6$: $(2, 3, 5, 7) \mapsto -\frac{753664}{17}i$, $(1, 2, 4, 8) \mapsto -\frac{4928}{15}i$, $(3, 4, 5, 6) \mapsto -762048i$, and crucially the lopsided $(2, 3, 100, 101)$ also matches — so it is not merely a “comparable magnitude” coincidence. At $n = 7$: $(2, 3, 5, 7, 11) \mapsto -\frac{4030464}{7}i$ and $(1, 2, 3, 4, 5) \mapsto -\frac{8896}{15}i$. At $n = 4$ the recursion is **Indeterminate** (the forced $0/0$), but the formula returns $-192i$ for $\{2, 3\}$ and $-40i$ for $\{1, 5\}$, matching the symbolic $n = 4$ limit $8i \omega_1 \omega_2^3 = -8i \omega_2^3 \omega_3$. Every test is exact.

Recovering the “ratio with a pole” the hint wants. If I keep the solved frequencies eliminated rather than substituted, the monomial becomes a genuine rational function of the free frequencies. With $S_1 = \omega_2 + \dots + \omega_{n-1} = -(\omega_1 + \omega_n)$ and $\omega_1 = -(S_1^2 - \omega_2^2 + \sum_{i \geq 3} \omega_i^2)/(2S_1)$,

$$A_n = -2^{n-2} i \frac{\omega_2^{2n-5} (S_1^2 - \omega_2^2 + \sum_{i=3}^{n-1} \omega_i^2)}{S_1}.$$

For $n = 5$ the bracket is $2(ab + b^2 + ac + bc + c^2)$, reproducing the chamber form; for $n = 6$ at $(2, 3, 5, 7)$ one has $S_1 = 17$ and bracket $= 289 - 4 + 83 = 368$, giving $-16i \cdot 128 \cdot 368/17 = -\frac{753664}{17}i$, as it must. The numerator does not vanish at $S_1 = 0$ (for $n = 5$ it reduces to $-\omega_3 \omega_4$), so the pole on the channel sub-energy S_1 is genuine — this is the “ratio of polynomials, simple channel-sub-energy pole, not a plain polynomial” the hint describes, now correctly understood as a per-chamber statement.

Trying for the fully general (any-chamber) rule. For robustness I attempt the generic $n = 5$ law. Writing $A_5 = 16i (\omega_1 \omega_2) \Phi$ with Φ a degree-four factor, the chambers above suggest that Φ

depends only on the two smallest magnitudes $f_1 \leq f_2$ and their signs s_1, s_2 (-1 for a minus leg):

$$\Phi = \begin{cases} f_1^4, & s_1 = -1, \\ f_1^2(2f_2^2 - f_1^2), & s_1 = +1, s_2 = -1, \\ 2f_1^2f_2^2, & s_1 = +1, s_2 = +1. \end{cases}$$

(I also note that both minus legs can never be the two smallest: the momentum law gives $\kappa_1 + \kappa_2 = \kappa_3 + \kappa_4 + \kappa_5$ with $\kappa_i = \omega_i^2$, and if κ_1, κ_2 were smallest then $\kappa_3 + \kappa_4 + \kappa_5 \geq 3 \max(\kappa_1, \kappa_2) > 2 \max \geq \kappa_1 + \kappa_2$, a contradiction. So the “ $s_1 = -1$ ” case always has $s_2 = +1$.) Testing this rule at 40 random points gives 38 exact matches; the two failures are instructive — one point has $\omega_5 = 0$ (a genuine degeneracy, recursion **Indeterminate**), and one has *negative* free frequencies, where a $(+, +)$ -smallest configuration disagrees ($-4.927 \times 10^6 i$ from the code versus $-5.690 \times 10^6 i$ from the rule). So once signs are mixed the chamber structure is finer than “the two smallest”, confirming that the genuinely global object is an irreducibly min / max / sorted thing. The clean single formula is the canonical-chamber one; the general case is the symmetric piecewise extension.

Why no single rational function can exist. I make the obstruction precise. The channel $\{2, 3, 4\}$ has $P_{234} = -\kappa_2 + \kappa_3 + \kappa_4$, and using $\kappa_3 + \kappa_4 = \kappa_1 + \kappa_2 - \kappa_5$ this is $P_{234} = \kappa_1 - \kappa_5$, whose sign is $\text{sgn}(|\omega_1| - |\omega_5|)$ — not fixed across the sector. So the unregularised $|k_S|$ genuinely flips, and the recursion is piecewise. Since A_5 equals the monomial $16i\omega_1\omega_2^5$ on the entire open chamber $\{|\omega_2| = \min\}$, any rational function agreeing with the code there must *be* that monomial, which disagrees in the adjacent chamber. Hence no single global rational function exists; the hint’s “one fit valid for all the sector” is achievable only within a chamber.

Choosing the chamber and verifying hard. The natural chamber is the one every example in the code inhabits: positive free frequencies in ascending order, so that ω_2 (the lone free minus leg, smallest of the free set) ends up the global minimum magnitude. I confirm $\omega_2 = \arg \min |\omega|$ holds at every such point. Against the *provided* **BGAmplitude** the boxed formula then matches *exactly* (relative error 0): 30/30 points at $n = 5$, 18/18 at $n = 6$, 8/8 at $n = 7$, plus explicit values like $\{3/2, 2, 5/2\} \mapsto -\frac{891}{2}i$ ($n = 5$), $\{1, 3/2, 2, 5/2\} \mapsto -\frac{968}{7}i$ ($n = 6$), $\{1, 3/2, 2, 5/2, 3\} \mapsto -\frac{1928}{5}i$ ($n = 7$).

An independent re-implementation as insurance. Because matching one code against a formula derived from it can hide a shared bug, I port the Berends–Giele recursion to Python from scratch — once in floating point and once in *exact* Gaussian-rational arithmetic (a small $a + bi$ class over the rationals). This separate codebase reproduces the Wolfram reference values $((2, 3, 5) \mapsto -3328i, (2, 3, 5, 7) \mapsto -\frac{753664}{17}i, (2, 3, 5, 7, 11) \mapsto -\frac{4030464}{7}i)$ and agrees *exactly* with the closed form over a fresh random sweep: 40/40 at $n = 5$, 20/20 at $n = 6$, 6/6 at $n = 7$. I also confirm the two forms of the answer (the all-frequency monomial and the free-frequency ratio) agree at 50 points each for $n = 4, 5, 6, 7$.

Closing $n = 4$ honestly. The recursion cannot evaluate $n = 4$ directly: the forced kinematics make $k_2 + k_4 = 0$ and $\omega_2 + \omega_4 = 0$, so the $\{2, 4\}$ propagator is a removable $0/0$ and the bare code returns **Indeterminate**. Taking the finite limit with $|\cdot|$ symbolic gives $A_4 = -8i\omega_2^3\omega_3 = 8i\omega_1\omega_2^3$ for $\omega_2 < \omega_3$ (and the swapped $-8i\omega_2\omega_3^3$ otherwise), exactly the formula; numeric limits $\{3/2, 5/2\} \mapsto -\frac{135}{2}i$, $\{1, 4\} \mapsto -32i$, $\{2, 7\} \mapsto -448i$ confirm it.

What the structure is telling me. The amplitude vanishing like ω_2^{2n-5} as the smallest frequency goes soft is a strong zero, not the single power one might expect, and it is the same phenomenon behind the $n = 4$ result $-8i\omega_2^3\omega_3$. The mechanism is that conservation pins the two minus legs to the extremes of the magnitude ordering with the plus legs in the middle, so that everything except the softest minus leg is fixed and that leg carries the remaining mass dimension by itself. The clean

statement is the monomial $A_n = 2^{n-1}i\omega_1\omega_2^{2n-5}$ (equivalently the displayed ratio with its S_1 pole), exact in the chamber selected by the standard ascending sampling; the genuinely global object is its symmetric, sorted-magnitude, piecewise extension, and no single rational function can represent it because the internal $|k_S|$ honestly changes sign within the sector.